

Wynton Zhang

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EDUCATION

Duke University | Durham, NC — Expected May 2028

Bachelor of Science in Engineering, Electrical and Computer Engineering

Minor: Machine Learning and Artificial Intelligence

Relevant coursework: Completed: Computer Architecture; Data Structures and Algorithms; Computer Network Architecture. In progress: Digital Systems; Robotics and Automation; Fields and Waves.

TECHNICAL PROJECTS

32-bit Processor Components | Verilog | ECE 350 — Fall 2026 · In progress

- Implemented a structural ALU with two-level carry-lookahead arithmetic, five-stage barrel shifters, bitwise logic, and signed overflow/comparison flags.
- Built a 32×32 register file with two read ports, one write port, and a hardwired zero register; checked simulation outputs against expected results using CSV-based testbenches.
- Developed a shift-and-add multiplier with a 64-bit accumulator, cycle counter, and result-ready control as an early checkpoint toward a complete processor.

Robot-Arm Planning and Manipulation | Python, ROS 2, MoveIt, Gazebo — Fall 2026

- Extended course scaffolds to implement Cartesian placement and multi-block stacking for a simulated Kinova Gen3 Lite arm, coordinating gripper release and collision-object attachment.
- Programmed obstacle-based trajectory comparisons, gripper-position checks, and a stop-on-missed-grasp recovery sequence.

EXPERIENCE

Software Engineering Intern | Duke | Durham, NC — Jun. 2025 – Aug. 2025

- Implemented a Variational Quantum Eigensolver in Qiskit and benchmarked results against Hartree–Fock and exact diagonalization.
- Compared classical optimizers and presented hybrid quantum–classical workflows to colleagues and Duke Quantum Information Society members.

Software Engineering Intern | Bluestamp | Aurora, CO — Jul. 2023 – Aug. 2024

- Developed Arduino and Raspberry Pi projects and an OpenCV computer-vision pipeline for robotics demonstrations; presented designs to an engineering cohort.

Product Management Intern | Greenhouse Scholars | Boulder, CO — Aug. 2024 – Present

- Developed data pipelines for scholar reporting and coordinated cross-functional teams to deliver student-engagement product features.

Electrical Engineering Intern | Eaglecrest High School | Arapahoe County, CO — Aug. 2022 – Apr. 2023

- Integrated LIDAR and ultrasonic sensors for object detection, created Matplotlib visualizations, and fabricated custom components for competition teams.

TECHNICAL SKILLS

Languages: Verilog, Python, Java, Bash

Tools & Platforms: Git, Icarus Verilog, GTKWave, Linux, Docker, Arduino, Raspberry Pi

Libraries & Robotics: NumPy, Pandas, Matplotlib, OpenCV, Qiskit, ROS 2, MoveIt, Gazebo

Design & Manufacturing: Onshape, Fusion, technical drawings, 3D printing, CNC milling